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Development of a Unified Banking Database System: Addressing Customer Information and Transactions

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I. Identified Data Tables

A. Customer Table

Attributes	Description
Customer_ID (PK)	Unique identifier (system-generated)
First_name	Customer's first name
Middle_name	Customer's middle name
Last_name	Customer's last name
Email	Customer's email address
Phone	Customer's phone number
Address	Customer's home address
Date_of_birth	Birthdate
Government_issued_ID	Valid ids issued by the government (eg. Postal ID, Passport, SSS ID, UMID, Voter's ID, etc.)

B. Account Table

Attributes	Description
Account_ID (PK)	Unique identifier (system-generated)
Customer_ID (FK)	Reference: Customer Table
Account_number	Unique identifier (system-generated)
Account_type	Savings, Checking, Loan
Balance	Customer's account balance
Currency	PHP, American Dollar, Yen, etc.
Status	Active, Dormant, Frozen
Creation_date	Date of account creation

C. Transaction Table

Attributes	Description
Transaction_ID (PK)	Unique identifier (system-generated)
Account_ID (FK)	Reference: Account Table
Target_Account_ID (FK)	Reference: Account Table (recipient, used for transfers)
Transaction_type	Deposit, Withdrawal, Transfer
Amount	Transaction amount
Currency	PHP, American Dollar, Yen, etc.
Transaction_status	Completed, Failed, Pending
Description	Brief note or memo for the transaction
Conf_num	Confirmation number (auto-generated)

D. Loans Table

Attributes	Description
Loan_ID (PK)	Unique identifier (system-generated)
Customer_ID (FK)	Reference: Customer Table
Account_ID (FK)	Reference: Account Table
Principal	Original loan amount borrowed
Interest_rate	Annual interest rate (e.g., 0.06 = 6%)
Month_terms	Loan duration in months
Status	Active, Paid, Defaulted
Start_date	Date the loan was issued
End_date	Expected loan end date
Loan_type	Personal, Housing, Auto, etc.

E. Audit Log

Attributes	Description
Log_ID (PK)	Unique identifier (system-generated)
Table_name	Name of the table modified (e.g., Customers, Account)
Record_ID	ID of the specific record that was changed
Action	Type of operation: Insert, Update, Delete
Changed_by	User or system that performed the action
Old_value	Previous value before the change
New_value	New value after the change
Changed_at	Timestamp of when the change occurred

II. Primary keys and Foreign keys Summary Table

	Primary Keys (PK)	Entity Name
1	Customer_ID	Customer
2	Account_ID	Account
3	Transaction_ID	Transaction
4	Loan_ID	Loans
5	Log_ID	Audit Log

	Foreign Keys (FK)	Entity Name
1	Customer_ID	Account
2	Account_ID	Transaction
3	Target_Account_ID	Transaction

4	Customer_ID	Loans
5	Account_ID	Loans

III. Entity Relationship Diagram (ERD)

Relationship and Cardinalities

